A3 Q2

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2. a) By the slides:

$\hat{Y} = PY$

and $r = Y - \hat{Y}$

$r = Y - PY$

$= IY - PY$

$= (I - P)Y$ by right distributivity

To show $(I-P)$ is also a projection matrix, we will show it is idempotent and symmetric.

Consider $(I-P)^T(I-P) = I^2 - P - P + P^2$

$= I - P - P + P$ P is idempotent

$= I - P$

This shows idempotence. And clearly, this matrix remains symmetric since I is diagonal and P is symmetric by def'n.

∴ We have shown that $r = (I-P)Y$ and that $(I-P)$ is a projection matrix.

b) Lin. combs. of X are norm. distr.

- Two components having Cov = 0 => they are indep.
  - If they are jointly norm. distr., $\rho = 0$ => they are indep.
- Mean $X = \mu$
- Cov $X = AA^T = \Sigma = \begin{bmatrix} \sigma_1^2 & \cdots & \sigma_{1n} \\ \vdots & \ddots & \\ \sigma_{n1} & & \sigma_n^2 \end{bmatrix}$ for $\rho = 1$
- $E[\hat{\beta}] = \beta$
- $\text{Cov}(\hat{\beta}) = \sigma^2(X^TX)^{-1} = \text{Var}(\hat{\beta})$
- $Y \sim MVN_n(X\beta, \sigma^2 I_n)$

We know that $\hat{\beta} \sim MVN(\beta, \sigma^2(X^TX)^{-1})$,

and that $\hat{Y} \sim MVN(X\beta, \sigma^2 P)$,

and that $Y \sim MVN(X\beta, \sigma^2 I)$

$r = Y - \hat{Y}$ where $\hat{Y} = PY$

$\text{Var}(r) = \text{Var}(Y) + \text{Var}(\hat{Y}) - 2\text{Cov}(Y, \hat{Y})$

$= \sigma^2 I + \sigma^2 P - 2\text{Cov}(IY, PY)$

$= \sigma^2 I + \sigma^2 P - 2[I\,\text{Cov}(Y)P^T]$

$= \sigma^2 I + \sigma^2 P - 2[\sigma^2 I P^T]$

$= \sigma^2 I + \sigma^2 P - 2\sigma^2 P$ (P is symmetric)

$= \sigma^2 I - \sigma^2 P$

$= \sigma^2(I - P)$

$E[r] = E[Y] - E[\hat{Y}]$

$= X\beta - X\beta$

$= 0$

$\therefore r \sim MVN(E[r], Var(r))$

$\equiv r \sim MVN(0, \sigma^2(I-\rho))$

Now, we find $E[U]$, $Cov(U)$

$E[U] = (E[\hat{\beta}], E[r])^T$

$= (\beta, 0)^T$

$Cov(U) = \begin{bmatrix} Var(\hat{\beta}) & Cov(\hat{\beta}, r) \\ Cov(\hat{\beta}, r) & Var(r) \end{bmatrix}$

$Cov(\hat{\beta}, r) = Cov((X^TX)^{-1}X^TY, (I-\rho)Y)$

$= (X^TX)^{-1}X^T Cov(Y)(I-\rho)^T$

$= (X^TX)^{-1}X^T\sigma^2 I(I - X(X^TX)^{-1}X^T)$ (drop ^{T} b/c in a), we showed $(I-\rho)$ is symmetric)

$= \sigma^2(X^TX)^{-1}X^T(I - X(X^TX)^{-1}X^T)$

$= \sigma^2(X^TX)^{-1}(X^T - X^TX(X^TX)^{-1}X^T)$

$= \sigma^2(X^TX)^{-1}(X^T - X^T)$

$= \sigma^2(X^TX)^{-1}(0)$

$= 0$

$\therefore U \sim MVN((E[\hat{\beta}], E[r])^T, Cov(\hat{\beta}, r))$

$\equiv U \sim MVN\left(\begin{bmatrix} \beta \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2(X^TX)^{-1} & 0 \\ 0 & \sigma^2(I-\rho) \end{bmatrix}\right)$

c) In b), we showed that $Cov(\hat{\beta}, r) = 0$ on the non-diagonal entries. Since their covariance is 0, by properties of MVN they are independent